ASSESSMENT TOOL 1
APPLICATION BASED QUESTIONS

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1. A factory has three boxes containing coloured balls. Box A contains 6 red, 4 blue, and 2 green balls; Box B contains 5 red, 7 blue, and 3 green balls; and Box C contains 8 red, 2 blue, and 5 green balls. One box is selected at random, but Box B is twice as likely to be chosen as each of the other boxes. A ball drawn from the chosen box is found to be blue. What is the probability that the selected box was Box C? (5 Marks)

Bayes' Theorem

Definition:

Bayes' Theorem is a fundamental concept in probability theory used to calculate the probability of an event based on prior knowledge and new evidence. It helps us update our prediction when additional information becomes available. In Machine Learning, Bayes' Theorem is widely used for classification, prediction, decision-making, and uncertainty analysis.

The theorem combines prior probability (initial belief) with likelihood (observed evidence) to determine the posterior probability (updated belief).

Learning Objective:

The objective of this problem is to understand how Bayes' Theorem updates the probability of an event after observing new evidence. This concept is important in Machine Learning because many intelligent systems make predictions by continuously updating probabilities based on incoming data.

Formula:

$$P(A \mid B) = \frac{P(B \mid A) \times P(A)}{P(B)}$$

Where:

- **P(A|B)** = Posterior Probability
- **P(B|A)** = Likelihood
- **P(A)** = Prior Probability
- **P(B)** = Total Probability

Given Data

Box	Red	Blue	Green	Total Balls	Selection Probability
A	6	4	2	12	1/4
B	5	7	3	15	1/2
C	8	2	5	15	1/4

Since Box B is twice as likely to be selected,

$$P(A) = \frac{1}{4}, P(B) = \frac{1}{2}, P(C) = \frac{1}{4}$$

Step 1: Calculate the Probability of Drawing a Blue Ball

For Box A,

$$P(Blue \mid A) = \frac{4}{12} = \frac{1}{3}$$

For Box B,

$$P(Blue \mid B) = \frac{7}{15}$$

For Box C,

$$P(Blue \mid C) = \frac{2}{15}$$

Step 2: Find the Total Probability of Drawing a Blue Ball

Using the Total Probability Rule,

$$P(Blue) = P(Blue \mid A)P(A) + P(Blue \mid B)P(B) + P(Blue \mid C)P(C)$$

Substituting the values,

$$= \left(\frac{1}{3} \times \frac{1}{4}\right) + \left(\frac{7}{15} \times \frac{1}{2}\right) + \left(\frac{2}{15} \times \frac{1}{4}\right)$$
$$= \frac{1}{12} + \frac{7}{30} + \frac{1}{30}$$

Taking LCM,

$$= \frac{5}{60} + \frac{14}{60} + \frac{2}{60}$$
$$= \frac{21}{60}$$
$$= \frac{7}{20}$$

Therefore,

$$P(Blue) = \frac{7}{20}$$

Step 3: Apply Bayes' Theorem

We need to find

$$P(Box \mid Blue)$$

Using Bayes' Theorem,

$$P(C \mid Blue) = \frac{P(Blue \mid C) \times P(C)}{P(Blue)}$$

Substituting the values,

$$= \frac{\left(\frac{2}{15} \times \frac{1}{4}\right)}{\frac{7}{20}}$$
$$= \frac{1}{30} \times \frac{20}{7}$$
$$= \frac{20}{210}$$
$$= \frac{2}{21} = 0.0952$$

Final Answer

$$\boxed{P(Box\ C \mid Blue) = \frac{2}{21} = 0.0952}$$

Therefore, the probability that the selected box was Box C is approximately 9.52%.

Interpretation:

The calculated probability shows that even though a blue ball was selected, it is not very likely that it came from Box C. This is because Box C contains only 2 blue balls out of 15, while Box B contains 7 blue balls out of 15 and also has a higher chance of being selected.

Reasoning:

This problem demonstrates conditional probability, where the probability of an event changes after new information becomes available. Initially, each box has a different probability of being selected. Once a blue ball is observed, the probability is updated using Bayes' Theorem. This approach helps make informed decisions under uncertainty, which is a key concept in Machine Learning and Artificial Intelligence.

Machine Learning Application:

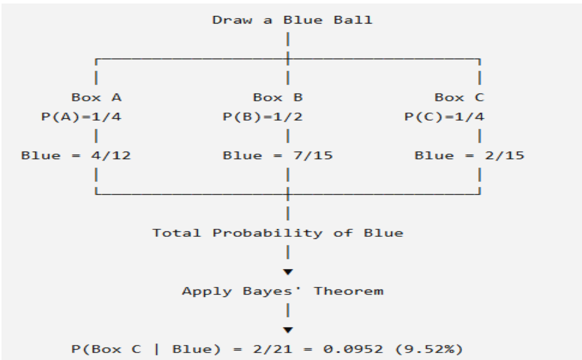
Bayes' Theorem is widely used in Machine Learning, especially in the Naïve Bayes Classifier. It is commonly used in:

- Spam Email Detection
- Medical Diagnosis
- Fraud Detection
- Weather Prediction.

Limitations (Critique)

- Depends on accurate prior probabilities.
- Incorrect assumptions may produce misleading results.
- Performance decreases if the input data is biased.
- Requires reliable historical data for better predictions.

Diagram:



Conclusion:

Bayes' Theorem is one of the most important concepts in probability and Machine Learning because it enables intelligent decision-making by updating probabilities based on new evidence. In this problem, although a blue ball was observed, the probability that it came from Box C is only 2/21 (9.52%) because Box B is more likely to be selected and contains a greater proportion of blue balls. This example demonstrates how Bayes' Theorem combines prior knowledge and observed data to make accurate predictions, making it an essential tool in fields such as Artificial Intelligence, healthcare, finance, cybersecurity, and predictive analytics.

2. A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be blue. What is the probability that the car came from Section C?

Introduction:

Bayes' Theorem is an important concept in probability that helps determine the probability of an event when new information is available. It combines prior knowledge with observed evidence to update the probability of an event. This theorem is widely used in Machine Learning, Artificial Intelligence, medical diagnosis, spam filtering, and decision-making systems. In this problem, Bayes' Theorem is used to determine the probability that a selected blue car came from Section C after observing that the chosen car is blue.

Definition:

Bayes' Theorem is a mathematical formula used to calculate the conditional probability of an event based on prior probability and new evidence.

Formula:

$$P(A \mid B) = \frac{P(B \mid A) \times P(A)}{P(B)}$$

Where:

- **P(A|B)** = Probability that event A occurs given event B has occurred.
- **P(B|A)** = Probability of event B occurring when A has occurred.
- **P(A)** = Prior probability of event A.
- **P(B)** = Total probability of event B.

Formula / Concept:

Bayes' Theorem is used to update the probability after observing new evidence.

$$P(C \mid Blue) = \frac{P(Blue \mid C) \times P(C)}{P(Blue)}$$

The Total Probability Rule is used to calculate the probability of selecting a blue car.

$$P(Blue) = P(Blue \mid A)P(A) + P(Blue \mid B)P(B) + P(Blue \mid C)P(C)$$

Given Data:

Section	Red	Blue	Black	Total Cars	Probability of Selection
A	5	3	2	10	1/4
B	4	6	5	15	1/2
C	7	2	6	15	1/4

Since Section B is twice as likely to be selected,

$$P(A) = \frac{1}{4}, P(B) = \frac{1}{2}, P(C) = \frac{1}{4}$$

Step-by-Step Solution

Step 1: Probability of selecting a blue car from each section

From Section A,

$$P(Blue \mid A) = \frac{3}{10}$$

From Section B,

$$P(Blue \mid B) = \frac{6}{15} = \frac{2}{5}$$

From Section C,

$$P(Blue \mid C) = \frac{2}{15}$$

Step 2: Calculate the total probability of selecting a blue car

$$P(Blue) = \left(\frac{3}{10} \times \frac{1}{4}\right) + \left(\frac{2}{5} \times \frac{1}{2}\right) + \left(\frac{2}{15} \times \frac{1}{4}\right)$$
$$= \frac{3}{40} + \frac{1}{5} + \frac{1}{30}$$

Taking LCM = 120,

$$= \frac{9}{120} + \frac{24}{120} + \frac{4}{120}$$
$$= \frac{37}{120}$$

Therefore,

$$P(Blue) = \frac{37}{120}$$

Step 3: Apply Bayes' Theorem

$$P(C \mid Blue) = \frac{P(Blue \mid C) \times P(C)}{P(Blue)}$$
$$= \frac{\left(\frac{2}{15} \times \frac{1}{4}\right)}{\frac{37}{120}}$$
$$= \frac{1}{30} \times \frac{120}{37}$$
$$= \frac{4}{37}$$
$$= 0.1081$$

Final Answer:

$$\boxed{P(\text{Section C} \mid \text{Blue}) = \frac{4}{37} = 0.1081}$$

Interpretation:

The probability of 10.81% indicates that it is less likely for the blue car to have come from Section C. This is because Section B has a higher chance of being selected and also contains more blue cars than Section C.

Reasoning:

Bayes' Theorem is suitable for this problem because it combines the prior probability of selecting each section with the probability of selecting a blue car from that section. This provides a more accurate result than simply looking at the number of blue cars. Such probability-based reasoning is commonly used in Machine Learning, data analysis, and intelligent decision-making systems.

Limitations (Critique)

- The result depends on the accuracy of the given probabilities.
- If the number of cars or selection probabilities change, the answer will also change.

- Bayes' Theorem assumes that the given information is correct and complete.

Diagram:

```
                 Parking Facility
                        |
      ┌─────────────────┼─────────────────┐
      |                 |                 |
  Section A         Section B         Section C
    P=1/4             P=1/2             P=1/4
      |                 |                 |
  Blue=3/10         Blue=6/15         Blue=2/15
      |                 |                 |
      └─────────────────┼─────────────────┘
                        |
              Apply Bayes' Theorem
                        |
                        ▼
    Probability (Section C | Blue) = 4/37
```

Conclusion:

Bayes' Theorem is a useful method for calculating conditional probability by combining prior information with observed evidence. In this problem, after observing that the selected car is blue, the probability that it came from Section C is 4/37 (approximately 10.81%). This example demonstrates how Bayes' Theorem can be applied to real-life decision-making problems and highlights its importance in probability, statistics, Machine Learning, and Artificial Intelligence.

3. In a real-world medical image classification task for detecting breast cancer, a deep-learning model integrated with the K-Nearest Neighbors algorithm is applied for binary classification (malignant vs. benign). The dataset consists of 1800 samples per class, with 20% of the data reserved for testing. During evaluation, the model reports 210 True Positives (TP), 190 True Negatives (TN), and 30 False Positives (FP). Based on the given information, determine the missing False Negatives (FN) value, and then calculate the Accuracy, Precision, Sensitivity (Recall), and Specificity of the classifier, explaining its effectiveness in medical diagnosis.

Medical Image Classification using K-Nearest Neighbors (KNN)

Given Data:

- Total samples = 1800 (Malignant) + 1800 (Benign) = 3600
- Testing data = 20% of 3600 = 720 samples

Given:

- True Positives (TP) = 210
- True Negatives (TN) = 190
- False Positives (FP) = 30
- False Negatives (FN) =?

Step 1: Calculate False Negatives (FN)

Total Test Samples = TP + TN + FP + FN

$$720 = 210 + 190 + 30 + FN$$

$$720 = 430 + FN$$

$$FN = 720 - 430$$

$$\boxed{FN = 290}$$

Step 2: Calculate Accuracy

Formula

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Substitute the values,

$$= \frac{210 + 190}{210 + 190 + 30 + 290}$$

$$= \frac{400}{720}$$

$$= 0.5556$$

$$\boxed{Accuracy = 55.56\%}$$

Step 3: Calculate Precision

Formula

$$Precision = \frac{TP}{TP + FP}$$

Substitute the values,

$$= \frac{210}{210 + 30}$$

$$= \frac{210}{240}$$

$$= 0.875$$

$$\boxed{Precision = 87.50\%}$$

Step 4: Calculate Sensitivity (Recall)

Formula

$$Sensitivity = \frac{TP}{TP + FN}$$

Substitute the values,

$$= \frac{210}{210 + 290}$$

$$= \frac{210}{500}$$

$$= 0.42$$

$$\boxed{Sensitivity = 42.00\%}$$

Step 5: Calculate Specificity

Formula

$$Specificity = \frac{TN}{TN + FP}$$

Substitute the values,

$$= \frac{190}{190 + 30}$$

$$= \frac{190}{220}$$

$$= 0.8636$$

$$\boxed{Specificity = 86.36\%}$$

Final Results

Performance Measure	Value
False Negatives (FN)	**290**
Accuracy	**55.56%**
Precision	**87.50%**
Sensitivity (Recall)	**42.00%**
Specificity	**86.36%**

Interpretation:

The classifier has a high precision (87.50%), indicating that when it predicts a patient has breast cancer, the prediction is usually correct. It also has a high specificity (86.36%), meaning it correctly identifies most healthy (benign) cases.

However, the sensitivity is only 42.00%, which means the model misses many actual malignant cases (high false negatives). In medical diagnosis, missing cancer cases can delay treatment and lead to serious consequences. Therefore, despite having good precision and specificity, this model is not sufficiently reliable for breast cancer screening because a higher sensitivity is essential to detect as many cancer cases as possible.

Conclusion:

The missing False Negatives (FN) value is 290. The classifier achieved 55.56% Accuracy, 87.50% Precision, 42.00% Sensitivity, and 86.36% Specificity. Although the model performs well in identifying healthy patients and making positive predictions accurately, its low sensitivity indicates that it fails to detect many malignant cases. For medical image classification, improving sensitivity is important to ensure early and accurate diagnosis of breast cancer.

4. Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs: (5 Marks)

(a) Epoch 1: Training Loss = 2.8, Validation Loss = 2.4
(b) Epoch 2: Training Loss = 2.5, Validation Loss = 2.2
(c) Epoch 3: Training Loss = 2.2, Validation Loss = 2.0
(d) Epoch 4: Training Loss = 2.0, Validation Loss = 2.1
(e) Epoch 5: Training Loss = 1.8, Validation Loss = 2.3
(f) Epoch 6: Training Loss = 1.6, Validation Loss = 2.6
(g) Epoch 7: Training Loss = 1.5, Validation Loss = 2.9
(h) Epoch 8: Training Loss = 1.3, Validation Loss = 3.2

i) Identify the epoch at which the model begins to overfit.
ii) Suggest two effective techniques to reduce overfitting in this model.
iii) Discuss how overfitting affects the model's generalization ability on unseen data.

Introduction:

Training and validation loss are important metrics used to evaluate the performance of a machine learning model. The training loss indicates how well the model learns from the training dataset, while the validation loss measures how well the model performs on unseen data. By comparing these values across different epochs, we can determine whether the model is learning effectively or beginning to overfit.

Definition:

- **Training Loss:** The error made by the model on the training dataset.
- **Validation Loss:** The error made by the model on the validation dataset, which is not used during training.
- **Overfitting:** A condition where the model performs well on training data but poorly on unseen data because it has memorized the training dataset instead of learning general patterns.

Learning Objective:

The objective of this problem is to analyse the training and validation loss values over multiple epochs, identify the point at which overfitting begins, suggest methods to reduce overfitting, and understand its effect on the model's performance

Given Data

Epoch	Training Loss	Validation Loss
1	2.8	2.4
2	2.5	2.2
3	2.2	2.0
4	2.0	2.1
5	1.8	2.3
6	1.6	2.6
7	1.5	2.9
8	1.3	3.2

Step-by-Step Analysis

(i) Identify the Epoch at Which the Model Begins to Overfit:

From Epoch 1 to Epoch 3, both the training loss and validation loss decrease, showing that the model is learning effectively.

At Epoch 4, the training loss continues to decrease (2.0), but the validation loss increases from 2.0 to 2.1.

This indicates that the model starts learning the training data too closely and its performance on unseen data begins to decrease. Therefore, the model begins to overfit at Epoch 4.

(ii) Two Techniques to Reduce Overfitting:

1. Early Stopping

Training should be stopped when the validation loss starts increasing. In this case, training can be stopped after Epoch 3 to prevent overfitting.

2. Regularization (Dropout or L2 Regularization)

Regularization reduces the complexity of the model by preventing it from memorizing the training data. Techniques such as Dropout randomly deactivate neurons during training, while L2 Regularization penalizes large weights, helping the model generalize better.

(iii) Effect of Overfitting on Generalization:

Overfitting reduces the model's ability to perform well on new or unseen data. Although the training loss continues to decrease, the validation loss increases, indicating that the model has memorized the training dataset instead of learning meaningful patterns. As a result, predictions on real-world data become less accurate and the model's generalization ability decreases.

Final Answer:

Question	Answer
Epoch where overfitting begins	**Epoch 4**
Technique 1	Early Stopping
Technique 2	Regularization (Dropout/L2 Regularization)
Effect of Overfitting	Reduces the model's ability to generalize and lowers prediction accuracy on unseen data.

Interpretation:
The training loss decreases continuously from 2.8 to 1.3, showing that the model is learning the training data. However, the validation loss decreases only until Epoch 3 and then starts increasing. This increasing validation loss is a clear sign of overfitting. Therefore, the best model is obtained around Epoch 3, before overfitting begins.

Conclusion:
The analysis shows that the machine learning model starts overfitting at Epoch 4, where the validation loss begins to increase while the training loss continues to decrease. Overfitting reduces the model's performance on unseen data and affects its generalization ability. Techniques such as Early Stopping and Regularization can effectively reduce overfitting and improve the overall performance of the model.

5. Examine the performance of a machine-learning model using training and validation accuracy over multiple epochs: (5 Marks)
(a) Epoch 1: Training Accuracy = 60%, Validation Accuracy = 58%
(b) Epoch 2: Training Accuracy = 65%, Validation Accuracy = 63%
(c) Epoch 3: Training Accuracy = 70%, Validation Accuracy = 68%
(d) Epoch 4: Training Accuracy = 75%, Validation Accuracy = 72%
(e) Epoch 5: Training Accuracy = 80%, Validation Accuracy = 73%
(f) Epoch 6: Training Accuracy = 85%, Validation Accuracy = 72%
(g) Epoch 7: Training Accuracy = 88%, Validation Accuracy = 70%
(h)Epoch 8: Training Accuracy = 92%,Validation Accuracy = 68%
i) Identify the epoch at which the model starts overfitting.
ii) Explain why validation accuracy decreases despite training accuracy increasing.
iii) Suggest two methods to improve validation performance.

Introduction:
Training accuracy and validation accuracy are important metrics used to evaluate the performance of a machine learning model. Training accuracy measures how well the model learns from the training data, while validation accuracy measures how well it performs on unseen data. Comparing these values across different epochs helps identify whether the model is learning effectively or starting to overfit.

Definition

- **Training Accuracy:** The percentage of correct predictions made by the model on the training dataset.
- **Validation Accuracy:** The percentage of correct predictions made by the model on the validation dataset.
- **Overfitting:** Overfitting occurs when a model performs very well on the training data but poorly on unseen data because it has memorized the training data instead of learning general patterns.

Learning Objective:

The objective of this problem is to analyse the training and validation accuracy over multiple epochs, identify when overfitting begins, explain the decrease in validation accuracy, and suggest methods to improve the model's performance on unseen data.

Given Data

Epoch	Training Accuracy	Validation Accuracy
1	60%	58%
2	65%	63%
3	70%	68%
4	75%	72%
5	80%	73%
6	85%	72%
7	88%	70%
8	92%	68%

Step-by-Step Analysis

(i) Identify the Epoch at Which the Model Starts Overfitting:

From Epoch 1 to Epoch 5, both training and validation accuracy increase, indicating that the model is learning effectively.

After Epoch 5, the training accuracy continues to increase, but the validation accuracy starts decreasing. Therefore, the model starts overfitting at Epoch 6.

(ii) Why Does Validation Accuracy Decrease While Training Accuracy Increases?

After Epoch 5, the model begins to memorize the training data instead of learning general patterns. As a result, the training accuracy keeps improving, but the model performs poorly on new or unseen validation data. This causes the validation accuracy to decrease, indicating that the model has started to overfit.

(iii) Two Methods to Improve Validation Performance:

1. Early Stopping

Stop the training process when the validation accuracy stops improving. In this case, the best model is obtained around **Epoch 5**.

2. Regularization (Dropout or L2 Regularization)

Regularization techniques reduce overfitting by preventing the model from becoming too complex. Dropout randomly deactivates neurons during training, while L2 Regularization penalizes very large weights, helping the model generalize better.

Final Answer:

Question	Answer
Epoch where overfitting starts	**Epoch 6**
Reason for decrease in validation accuracy	The model memorizes the training data and fails to generalize to unseen data.
Method 1	Early Stopping
Method 2	Regularization (Dropout/L2 Regularization)

Interpretation:

The training accuracy increases steadily from 60% to 92%, showing that the model is learning the training data well. The validation accuracy also increases until Epoch 5, reaching 73%, but then decreases from 72% to 68%. This indicates that the model begins to overfit after Epoch 5. The best-performing model is therefore obtained around Epoch 5, before overfitting starts.

Conclusion

The machine learning model begins overfitting at Epoch 6, where the validation accuracy decreases while the training accuracy continues to improve. This shows that the model is learning the training data too closely and losing its ability to generalize to unseen data. Using Early Stopping and Regularization can reduce overfitting, improve validation performance, and produce a more reliable model for real-world applications.